

MTH-210: End-Semester Model Solution

Monsoon-2025

December 8, 2025

Full Marks: 70

Time: 3:00 hours

General Instructions: This is a closed book examination. Please be precise, brief and to the point in your answer. Unnecessary blabbers or unreadable handwriting will fetch deduction of marks. Also, if you want to disprove a statement by a counterexample, then your counterexample must be explicit. In addition, your counterexample must be complete and provide a clear explanation why it is a counterexample. Vague ideas and reasonings will not incur full credits.

1. Let B be a set and S_1 and S_2 be two distinct equivalence relations on B . Define $S_1 \oplus S_2 = (S_1 \cup S_2) - (S_1 \cap S_2)$. Then, prove or disprove the following statements.

- (a) $S_1 \oplus S_2$ is an equivalence relation.
(b) $S_1 \oplus S_2$ is a symmetric relation.

(5 Marks + 5 Marks)

Solution:

- (a) The statement is **FALSE**.

Counter Example: Consider a set $B = \{a, b, c, d\}$ and two relations

- $S_1 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b), (a, c), (c, a)\}$ and
- $S_2 = \{(a, a), (b, b), (c, c), (d, d), (b, c), (c, b), (c, d), (d, c), (b, d), (d, b)\}$.

Observe that S_1 and S_2 are equivalence relations. Then, $S_1 \oplus S_2 = \{(a, b), (b, a), (c, d), (d, c), (b, d), (d, b)\}$. Clearly, $S_1 \oplus S_2$ is not reflexive. Hence, $S_1 \oplus S_2$ is not an equivalence relation.

- (b) The statement is **TRUE**.

Proof: Since S_1 and S_2 are two distinct equivalence relations (by assumption), both S_1 and S_2 are symmetric relations.

Now, consider any $(x, y) \in S_1 \oplus S_2$.

Case-(i): $(x, y) \in S_1$ but $(x, y) \notin S_2$.

Since S_1 is symmetric, therefore, $(y, x) \in S_1$. As $(x, y) \notin S_2$ and S_2 is symmetric, it follows that $(y, x) \notin S_2$.

It implies that $(y, x) \in S_1 \oplus S_2$.

Case-(ii): $(x, y) \in S_2$ but $(x, y) \notin S_1$.

By similar arguments as Case-(i) we can prove that $(x, y) \in S_2$ but $(x, y) \notin S_1$.

Then, $(x, y) \in S_1 \oplus S_2$.

As the cases are exhaustive, this completes the proof that $S_1 \oplus S_2$ is a symmetric relation.

1 mark for proving/disproving in each part.

a) 1 mark for showing S_1 is an equivalence relation
1 mark for showing S_2 is an equivalence relation
2 marks for showing $S_1 \oplus S_2$ is not an equivalence relation

2. Let S_1 be a partial order on A and S_2 be a partial order on B . Define a relation S on $A \times B$ as follows: $((a, b), (c, d)) \in S$ if and only if $(a, c) \in S_1$ and $(b, d) \in S_2$.

Prove or disprove: if S_1 is a total order on A and S_2 is a total order on B , then S is a total order on $A \times B$. (8 Marks)

Solution: The statement is **FALSE**.

1 mark

Counter example: Consider $A = \mathbb{N}$ and $B = \mathbb{N}$.

Consider $S_1 = \{(a, b) \mid a, b \in \mathbb{N}, a \leq b\}$ and $S_2 = S_1$.

Then, by definition $S = \{((a, b), (c, d)) \mid a, b, c, d \in \mathbb{N}, a \leq c \text{ and } b \leq d\}$.

If S is a total order on $\mathbb{N} \times \mathbb{N}$, then for every two distinct elements $(x_1, y_1), (x_2, y_2)$, it must be that $((x_1, y_1), (x_2, y_2)) \in S$ or $((x_2, y_2), (x_1, y_1)) \in S$.

Now, we consider $(7, 3)$ and $(1, 5)$ both are elements of $\mathbb{N} \times \mathbb{N}$.

Observe that $3 < 5$ but $7 > 1$.

Similarly, $1 < 7$ but $5 > 3$.

That is why $((7, 3), (1, 5)), ((1, 5), (7, 3)) \notin S$.

Hence, $(7, 3)$ and $(1, 5)$ are not comparable.

Therefore, S is not a total order.

2 marks for showing total order on A.

2 marks for showing total order on B.

3. Let B be the set of all real numbers that do not have 2 in their decimal representation. Then, prove or disprove that B is countable. (8 Marks)

Solution: This statement is **FALSE**.

1 mark

This also needs to be **PROVED BY DIAGONALIZATION ARGUMENT**.

Proof: We consider a subset of B which is defined as

$D = \{x \in \mathbb{R} \mid 3 < x < 4 \text{ and } x \text{ does not have 2 in their decimal representation}\}$.

Note that $D \subseteq B$ and observe that $D \subseteq (3, 4)$ where $(3, 4)$ is the open interval between 3 and 4 in the real-line.

This is proof by contradiction.

Suppose that D is countable. Then, there exists a bijection $g : \mathbb{N} \rightarrow D$.

For every $k \in \mathbb{N}$, we represent $g(k)$ by $3.a_{k,0}a_{k,1} \dots a_{k,k} \dots$.

In short, $a_{k,i}$ denotes the $(i+1)$ -th digit of $g(k)$ after the decimal point.

We construct a number w the digits of which are represented as $3.b_0b_1b_2 \dots$ as follows.

For every $k \in \mathbb{N}$, if $a_{k,k} \neq 4$, then we set $b_k = 4$.

Otherwise, $a_{k,k} = 4$, we choose $b_k = 5$.

In short summary, from $f(k)$, we modify the $(k+1)$ -th digit after decimal point for $g(k)$.

As we do not choose 0 or 9 for making this change, it follows that w is different from $g(k)$ for every $k \in \mathbb{N}$.

But w does not have 2 in its decimal representation.

Therefore, $w \in D$ but $w \neq g(k)$ for every $k \in \mathbb{N}$.

Hence, there exists $w \in D$ that has no pre-image under g of any $k \in \mathbb{N}$.

It implies that g is not surjective. This contradicts that g is a bijection.

Therefore, D is uncountable.

As D is a subset of B and D is uncountable, therefore B is also uncountable.

1 mark

2 marks for using result & explanation.

4. A relation S on a set B is *circular* if $(a, b) \in S$ and $(b, c) \in S$, then $(c, a) \in S$. Prove or disprove: if S is circular and reflexive, then S is an equivalence relation. (8 Marks)

Solution: This statement is **TRUE**.

1 mark

Proof: Let S be a relation on a set B that is circular and reflexive.

By assumption, S is a reflexive relation.

1 mark for reflexivity

Now, consider $(x, y) \in S$.

As S is reflexive, $(y, y) \in S$.

By assumption, S is circular.

Then, $(y, x) \in S$.

Therefore, $(y, x) \in S$ whenever $(x, y) \in S$, implying that S is symmetric.

Next, assume that $(x, y), (y, z) \in S$.

As S is circular, $(z, x) \in S$.

We have already proved that S is symmetric.

Therefore, due to $(z, x) \in S$, it implies that $(x, z) \in S$.

Therefore, if $(x, y), (y, z) \in S$, then $(x, z) \in S$.

Hence, S is transitive.

As S is reflexive, symmetric, and transitive, it follows that S is an equivalence relation.

3 for symmetric

3 for transitive

5. In the xy -plane, consider two coordinates $(0, 0)$ and (m, n) for some integers $m, n > 0$. A path from $(0, 0)$ to (m, n) is a *lattice path* if the path is made up of a series of steps, where each step is a move one unit to the right or a move one unit upward. No move to the left or downward is allowed. Explain by using a counting argument how many lattice paths are there from $(0, 0)$ to (m, n) . (6 Marks)

Solution: Every lattice path is a path from $(0, 0)$ to (m, n) . Observe that, any lattice path must have exactly $m + n$ steps. Out of these $m + n$ steps, each of m steps must be one unit to the right, and each of other n steps must be one unit to the upward.

We represent a lattice path as a string over $\{0, 1\}$ of $m + n$ bits as follows.

If a step moves one unit to the right, then we represent that step as bit 1. Otherwise, a step moves one unit to the upward. We represent this step as bit 0.

It is clear that there must be exactly m many 1s and n many 0s. But those 1s and 0s can appear in any bit position out of these $m + n$ distinct bit positions.

Every distinct position, indexed by $1, 2, \dots, m + n$ forms a set $A = \{1, 2, \dots, m + n\}$.

Additionally, the set of positions in which the bit is 1 represents a subset of A that has length exactly m .

Hence, for every bit string of length n over $\{0, 1\}$ is represented by a subset of m elements out of a set of $m + n$ elements.

Therefore, there are $\binom{m+n}{m}$ distinct lattice paths from $(0, 0)$ to (m, n) .

6. Let n and k be positive integers such that $n \geq k$. Prove by using combinatorial argument that $\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$. (8 Marks)

[N.B. If you prove by algebraic manipulation, then you will be awarded zero marks. Your proof must be double-counting argument or bijective proof]

Solution: We prove this by using a double-counting argument.

4 marks for explanation

2 marks for final answer

4 marks for explanation of LHS.
4 marks for explanation of RHS.

Consider a set M of n professors in Mathematics department and a set C of n professors of Computer Science (CS) department.

We represent the LHS by a committee $B \subseteq M \cup C$ of exactly n members that must satisfy two requirements.

- at least 1 member of the committee B must be from M , and
- the chairperson of B must be from M .

To construct such a committee, what we can do is the following. We have to first select a set of k professors from M such that $1 \leq k \leq n$. In addition, we choose 1 chairperson out of these k chosen professors from M . For every fixed $k \in \{1, 2, \dots, n\}$, this selection can be performed in $k \binom{n}{k}$ ways. This is the **FIRST PART** of counting

Subsequently, for each such $k \in \{1, 2, \dots, n\}$ and the choice under the first part, we choose $n - k$ remaining professors from the CS department. This can be achieved in $\binom{n}{n-k}$ ways.

We use that $\binom{n}{n-k} = \binom{n}{k}$.

Then, the total number of ways to form a committee of n members with the chairperson being from Maths department is

$$\sum_{k=1}^n k \binom{n}{k} \binom{n}{k}^2$$

this is exactly the LHS.

Now, we consider the RHS.

First, we choose a mathematics professor who is the chair of this committee B . This can be achieved in n ways.

After choosing the chairperson, we choose the other $n - 1$ professors in the committee B from the remaining $2n - 1$ professors in $M \cup C$. This committee will have n professors from $M \cup C$ such that the chairperson must be from Mathematics department. This can be achieved in $\binom{2n-1}{n-1}$ ways.

Therefore, the total number of ways to form a committee of n professors from $M \cup C$ with chairperson from Mathematics is $n \binom{2n-1}{n-1}$. This is exactly the RHS.

Therefore, LHS = RHS.

7. If n and d are positive integers, then $n \pmod{d}$ denotes the remainder when n is divided by d . Prove or disprove that: if a set of 20 ordered pairs are chosen from $\mathbb{Z}^+ \times \mathbb{Z}^+$, then there are 3 distinct sets $\{(a_1, a_2), (b_1, b_2)\}$, $\{(p_1, p_2), (q_1, q_2)\}$, and $\{(d_1, d_2), (r_1, r_2)\}$ such that

- $a_1 \pmod{3} = b_1 \pmod{3}$, $a_2 \pmod{3} = b_2 \pmod{3}$,
- $p_1 \pmod{3} = q_1 \pmod{3}$, $p_2 \pmod{3} = q_2 \pmod{3}$, and
- $d_1 \pmod{3} = r_1 \pmod{3}$, $d_2 \pmod{3} = r_2 \pmod{3}$.

1 mark for attempt of proving (8 Marks)

[N.B. $37 \pmod{5} = 2$ because if 37 is divided by 5, then we get that the quotient is 7 and the remainder is 2.]

Solution: Consider an arbitrary ordered pair $(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+$.

The value of $x \pmod{3} \in \{0, 1, 2\}$. Similarly, the value of $y \pmod{3} \in \{0, 1, 2\}$.

Therefore, there are 3 choices for the value of $x \pmod{3}$ and for each such possibility 3 choices for the value of $y \pmod{3}$.

By product rule, there are 9 possible choices for the values $(x \pmod{3}, y \pmod{3})$.

When 20 distinct ordered pairs are chosen from $\mathbb{Z}^+ \times \mathbb{Z}^+$, due to PIGEON-HOLE PRINCIPLE, there are $\lceil \frac{20}{9} \rceil$ such ordered pairs, hence 3 such ordered pairs (a_1, b_1) , (a_2, b_2) , (a_3, b_3) that satisfy the following properties.

2 marks for application of Pigeon-Hole Principle correctly (No statement needed).

2 marks for constructing this $\{0, 1, 2\} \times \{0, 1, 2\}$ with explanation.

- $a_1 \pmod{3} = a_2 \pmod{3} = a_3 \pmod{3}$, and
- $b_1 \pmod{3} = b_2 \pmod{3} = b_3 \pmod{3}$.

Then, consider $S = \{(a_1, b_1), (a_2, b_2), (a_3, b_3)\}$.

There are $\binom{3}{2}$ ways, hence 3 ways to choose a set of size 2 from S .

These sets are $\{(a_1, b_1), (a_2, b_2)\}$, $\{(a_2, b_2), (a_3, b_3)\}$, and $\{(a_1, b_1), (a_3, b_3)\}$.

Observe that in these above mentioned sets, the following properties are satisfied.

- $a_1 \pmod{3} = a_2 \pmod{3}$, $b_1 \pmod{3} = b_2 \pmod{3}$,
- $a_2 \pmod{3} = a_3 \pmod{3}$, $b_2 \pmod{3} = b_3 \pmod{3}$, and
- $a_1 \pmod{3} = a_3 \pmod{3}$, $b_1 \pmod{3} = b_3 \pmod{3}$.

Therefore, the claimed statement is true.

8. Let $G = (V, E)$ be a simple graph with n vertices and G has an edge whose removal makes it a tree. Let V_1 denote the set of all pendant vertices, V_2 denote the set of all vertices with degree 2, and V_3 denote the vertices of G with degree at least 3. Then, prove or disprove that $|V_3| \leq |V_1|$. **(7 Marks)**

Solution: As G is a simple graph with n vertices, and there exists one edge the removal of which makes it a tree, the first crucial observation is that G has exactly n edges.

Because, it is well-known due to a lemma (taught in class) that a tree with n vertices has exactly $n - 1$ edges. Hence, G has exactly n edges.

A vertex is a pendant vertex if it has degree exactly one.

Hence, the following is true

$$\sum_{v \in V(G)} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v) + \sum_{v \in V_3} \deg(v)$$

Observe that for every $v \in V_1$, $\deg(v)$ counts to exactly 1. Hence, $\sum_{v \in V_1} \deg(v) = |V_1|$.

Similarly, for every $v \in V_2$, $\deg(v)$ counts to exactly 2. Hence, $\sum_{v \in V_2} \deg(v) = 2|V_2|$.

For every $v \in V_3$, $\deg(v) \geq 3$. Therefore, what is crucial is that $\sum_{v \in V_3} \deg(v) \geq 3|V_3|$.

As $\sum_{v \in V} \deg(v) = 2n$ and combining the above 3 facts, we get that

$$2n = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v) + \sum_{v \in V_3} \deg(v) \geq |V_1| + 2|V_2| + 3|V_3|$$

Now, observe that V_1, V_2 , and V_3 are pairwise disjoint. As G is connected and G has no isolated vertices, it follows that $|V_1| + |V_2| + |V_3| = n$.

Then, $2|V_1| + 2|V_2| + 2|V_3| \geq |V_1| + 2|V_2| + 3|V_3|$.

Then, $|V_1| \geq |V_3|$.

Therefore, the statement is true.

9. Let $f : A \rightarrow B$ be a well-defined function. For a subset $P \subseteq A$, $f(P) = \{y \in B \mid y = f(x) \text{ for some } x \in P\}$. Then, prove or disprove: for every $D_1, D_2 \subseteq A$, $f(D_1) \cap f(D_2) = f(D_1 \cap D_2)$. **(7 Marks)**

Solution: This statement is **FALSE**.

1 mark

Counterexample: Consider $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4\}$.

Define $f(a) = 1$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$.

Consider $D_1 = \{a, b\}$ and $D_2 = \{b, c, d\}$.

Then, $f(D_1) = \{1, 2\}$ and $f(D_2) = \{1, 2, 3\}$.

Observe that $D_1 \cap D_2 = \{b\}$.

Then, $f(D_1 \cap D_2) = f(b) = \{2\}$.

But, $f(D_1) \cap f(D_2) = \{1, 2\}$.

Therefore, this statement is false.

2 marks for
defining function.
1 mark for writing
 D_1, D_2

3 marks for explanation.

APPENDIX:

- The set $\mathbb{Z}^+$ denotes the set of all positive integers which is the same set as $\mathbb{N} - \{0\}$. Equivalently, $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$.
- If you want to prove that an infinite set is countable, then please prove the existence of a precise bijection. If you want to prove that an infinite set is uncountable, then you can use diagonalization and some properties of uncountable sets.
- An undirected graph is a *simple graph* if it has no self loop or parallel edges.
- An undirected graph is *connected* if there is a simple path between every pair of vertices.
- A *cycle* is a simple path that starts and ends at the same vertex.
- A *tree* is an undirected graph that is connected but has no cycle.
- You can use any other lemma/theorem that was taught in the class.

NOTE :

1.

Aditya

Q. 1

Sourav.

Q. 2, Q. 8

Mahesh

Q. 3, Q. 6

Asmit

Q. 4

Bikram

Q. 5

Aahan

Q. 7

Esha

Q. 9

2. In questions 1, 2, 3, 4, 7, 8, 9, if students fail to deduce true/false correctly, they will get zero marks.

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